AgonProject

An online platform for exploring approaches to formal argumentation

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Why AgonProject?

Formal argumentation offers a rich *zoo* of formalisms for reasoning with conflicting information — but their diverse semantics, algorithms, and implementation details make them hard for students and newcomers to grasp.

TweetyProject provides the most comprehensive collection of argumentation implementations available, yet exposes only *text-based* interfaces for a handful of tasks.

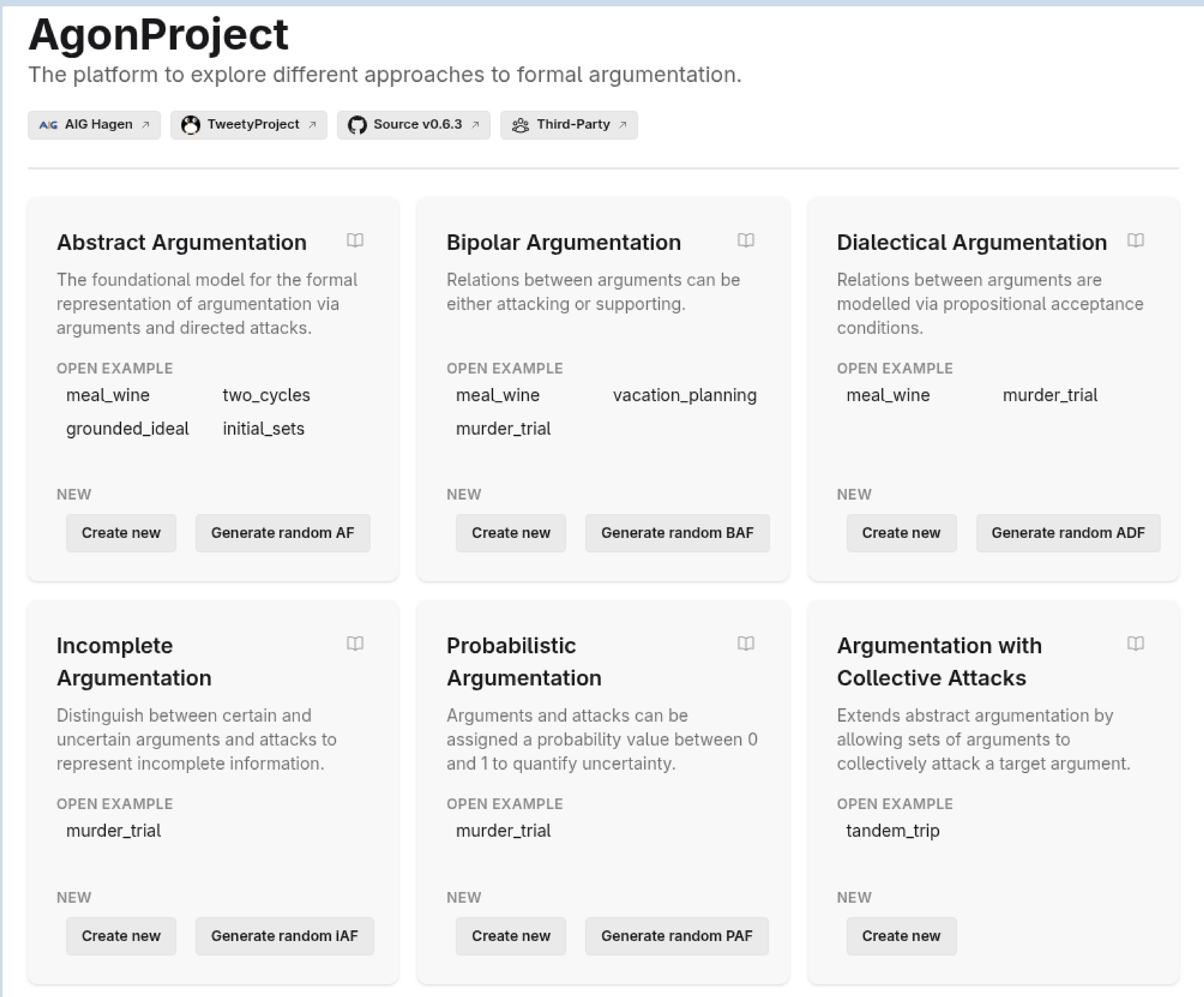
Goal: a unified, **visual**, install-free web interface to *construct*, *evaluate*, and *compare* argumentation formalisms — right in the browser.

Built with Vue 3 + TypeScript (client-side single-page app); reasoning tasks are delegated to a TweetyProject REST backend. Open source under GPL-3.0.

Try it now — no installation

agon.tweetyproject.org

One Platform — Six Argumentation Formalisms



AF *Abstract* argumentation — arguments and directed attacks (Dung).

BAF *Bipolar* — attacks **and** a support relation.

ADF *Abstract dialectical* — propositional acceptance conditions.

iAF *Incomplete* — uncertain arguments and attacks.

PAF *Probabilistic* — probabilities on arguments and attacks.

SetAF *Collective attacks* — sets of arguments attack a target.

Every formalism shares the same editor, evaluation, export, generation, and learning features below — so comparing them is effortless.

What You Can Do

Build
Interactive graph editor with directed, circular, radial and force-directed layouts (Graphviz) plus a physics simulation.

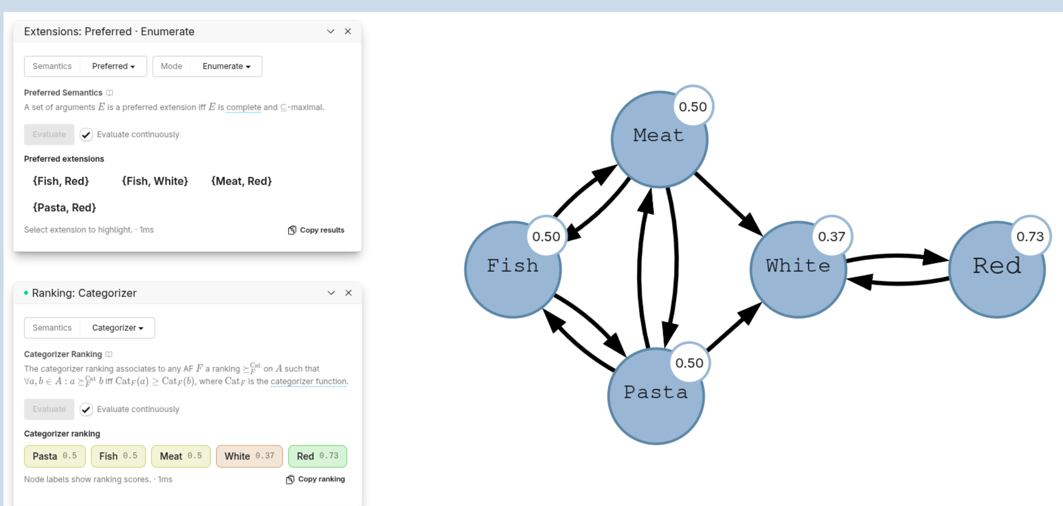
Evaluate
Extension- and ranking-based semantics; credulous / sceptical acceptance; highlight extensions directly in the graph; compare in parallel windows.

Export & Share
One-click LaTeX (`argumentation` package), SVG, TGF, and ICCMA AF-format — plus a shareable URL.

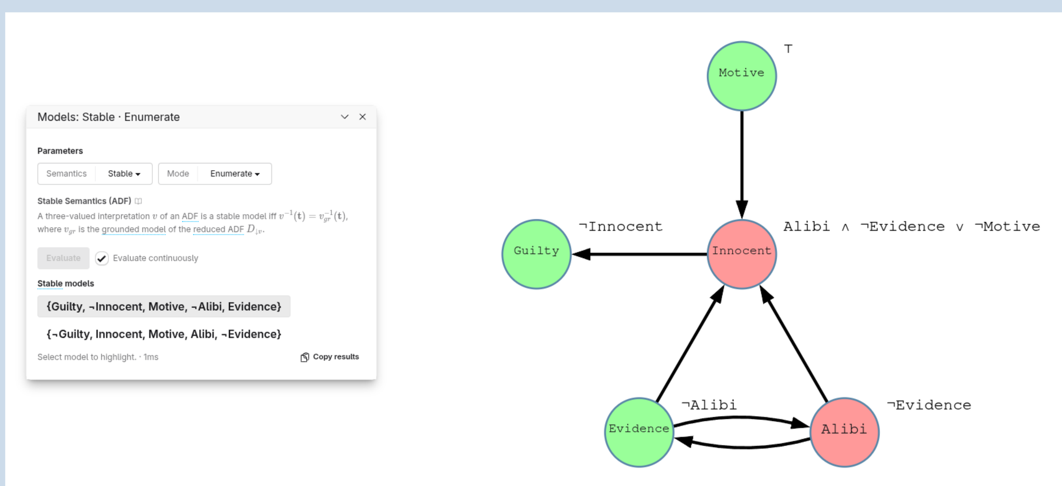
Generate
Random frameworks from eight `networkX` generators (Erdős–Rényi, Watts–Strogatz, Barabási–Albert, . . .).

Learn
10+ step-by-step tutorials, an extensive glossary with literature references, and curated example frameworks.

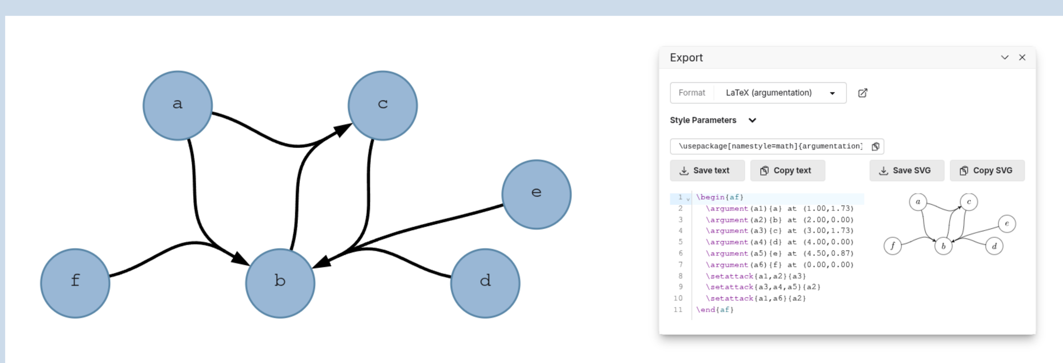
In Action



Evaluate & rank (AF). Preferred extensions are enumerated while the *categoriser* ranking is annotated directly on each argument; evaluation windows can stay open in parallel.



Acceptance conditions (ADF). Each argument carries a propositional acceptance condition; AgonProject evaluates three-valued interpretations and highlights a stable model in the graph.



Collective attacks & export (SetAF). A *set* of arguments jointly attacks a target, drawn as hyperedges; the export window emits ready-to-use LaTeX with an SVG preview.

Supported Semantics

Group	Semantics
Classical	conflict-free, admissible, complete, grounded, preferred, stable
Admissibility-based	ideal, semi-stable, eager, strong admissibility, initial sets, unchallenged
Non-admissible	naive, CF2, SCF2, stage, stage2, (strongly) undisputed
Weak-admissible	w-admissible, w-complete, w-grounded, w-preferred
Rankings	burden, categoriser, counting, discussion, iterated graded defense, serialisability, social, strategy, tuples

Available across AF, BAF, iAF, PAF and SetAF; ADFs use the three-valued classical and naive semantics. Serialisation sequences are supported interactively.

Outlook

Planned extensions:

- further formalisms — Assumption-Based Argumentation, Weighted AFs, Bipolar Set-based AFs, claim-augmented AFs;
- benchmark generators from the ICCMA ecosystem;
- explanation-based reasoning: *why* is an argument accepted?

AgonProject is backend-agnostic and open to new formalisms and solvers.

Live: agon.tweetyproject.org • **Source:** github.com/aig-hagen/AgonProject • GPL-3.0

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LaTeX TikZposter